

# Python Basics Guide

## 1. What is Python?

Python is a high-level, interpreted programming language known for its simple and readable syntax. It is widely used in web development, data science, automation, artificial intelligence, and more.

## 2. Hello World Program

The first program most programmers write prints Hello World.

Example: `print('Hello World')`

## 3. Variables

Variables are used to store data in memory.

Example:

```
name = 'Nikhil'
```

```
age = 21
```

## 4. Data Types

- 1 int - Integer numbers
- 2 float - Decimal numbers
- 3 str - Text values
- 4 bool - True or False
- 5 list, tuple, dictionary - Collection types

## 5. Conditional Statements

Used to make decisions in programs.

Example:

```
if age >= 18:  
    print('Adult')  
else:  
    print('Minor')
```

## 6. Loops

Loops repeat code multiple times.

For Loop Example:

```
for i in range(5): print(i)
```

While Loop Example:

```
while x < 5: x += 1
```

## 7. Functions

Functions group reusable code.

Example:

```
def greet(name):  
    print('Hello', name)
```

## 8. Conclusion

You now know the fundamental basics of Python programming. Practice daily to improve your skills!